

MH3100 Real Analysis I

Tutorial 2 Solutions

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Overview

- **Lecture coverage:** Lectures 4–6 — the definition of the limit of a sequence, the Algebraic Limit Theorem, the Order Limit Theorem, the Squeeze Theorem, and Cauchy sequences.
- **Questions:** 8 questions from Tutorial Question Set No. 2 (main Q1–Q5, Additional Questions A1–A3).
- **Notation:** as in lectures — $\mathbb{N} = \{1, 2, 3, \dots\}$ (does *not* include 0); a sequence is $\{a_n\}$; a limit is written value-first, $a = \lim a_n$; a property holding eventually means it holds on some tail $a_N, a_{N+1}, \dots$.
- **Running restriction:** Q4 and Q5 explicitly forbid the Algebraic Limit Theorem and (for Q5) the Cauchy Criterion; every proof in those two questions is therefore built from the raw ε - N definitions only.

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Question 1 (Eight true/false statements about convergence and divergence)

Problem

Determine if each of the following statements is true.

- (a) There is a sequence with an infinite number of 1's and does not converge to 1.
- (b) There is a sequence with an infinite number of 1's and converges to a limit not equal to 1.
- (c) There is a divergent sequence such that for every $n \in \mathbb{N}$ it is possible to find n consecutive ones somewhere in the sequence.
- (d) There are sequences $\{a_n\}$ and $\{b_n\}$, which both diverge, but whose sum $\{a_n + b_n\}$ converges.
- (e) There are sequences $\{a_n\}$ and $\{b_n\}$, where $\{a_n\}$ converges, $\{b_n\}$ diverges, and $\{a_n + b_n\}$ converges.
- (f) There is a convergent sequence $\{b_n\}$ with $b_n \neq 0$ for each $n \in \mathbb{N}$ such that $\{1/b_n\}$ diverges.
- (g) There is an unbounded sequence $\{a_n\}$ and a convergent sequence $\{b_n\}$ with $\{a_n - b_n\}$ bounded.
- (h) There are sequences $\{a_n\}$ and $\{b_n\}$, where $\{a_n b_n\}$ and $\{a_n\}$ converge but $\{b_n\}$ does not.

(Tutorial 2, Q1.)

Solution

Method 1: examples; contradiction from the definition

(a) True. $a_n := 1$ (n odd), $a_n := 0$ (n even); $a_n = 1$ for infinitely many n . Take $\varepsilon = \frac{1}{2}$: for every $N \in \mathbb{N}$, $n := 2N \geq N$ and $a_n = 0$, so $|a_n - 1| = 1 \not< \frac{1}{2}$. $\therefore \{a_n\}$ does not converge to 1.

True; $a_n = 1$ (n odd), $a_n = 0$ (n even).

(b) False.

Proof. Suppose $a_n = 1$ for infinitely many n and $a = \lim a_n$ with $a \neq 1$. Put $\varepsilon := \frac{1}{2}|a - 1| > 0$. There is N with $|a_n - a| < \varepsilon$ for all $n \geq N$; pick $n_0 \geq N$ with $a_{n_0} = 1$.

$$|a - 1| = |a_{n_0} - a| < \frac{1}{2}|a - 1| \implies |a - 1| < 0 \quad (\text{contradiction})$$

$\therefore$ no such limit $a \neq 1$ exists. □

False; a sequence with infinitely many 1's can only converge to 1, never to another limit.

(c) **True.** $\{a_n\} : 1, 0, 1, 1, 0, 1, 1, 1, 0, \dots$ (a run of k ones, then one 0, for $k = 1, 2, 3, \dots$). Run $k = n$ gives n consecutive 1's; both 0 and 1 occur infinitely often. Suppose $a = \lim a_n$; take $\varepsilon = \frac{1}{2}$: there is N with $|a_n - a| < \frac{1}{2}$ for all $n \geq N$. Pick $n_1, n_2 \geq N$ with $a_{n_1} = 0$, $a_{n_2} = 1$.

$$1 = |a_{n_2} - a_{n_1}| \leq |a_{n_2} - a| + |a - a_{n_1}| < \frac{1}{2} + \frac{1}{2} = 1 \quad (\text{contradiction})$$

$\therefore \{a_n\}$ diverges.

True; growing runs of 1's (length n) separated by single 0's, diverging by the argument above.

(d) **True.** $a_n := n$, $b_n := -n$. For every $M > 0$ there is $n > M$ (Archimedean), so $|a_n| = |b_n| > M$. $\therefore \{a_n\}, \{b_n\}$ are unbounded, hence divergent (convergent $\implies$ bounded). $a_n + b_n = 0$ for all $n \therefore \{a_n + b_n\}$ converges to 0.

True; $a_n = n$, $b_n = -n$, $a_n + b_n = 0$ for all n .

(e) **False.**

Proof. Suppose $a = \lim a_n$, $s = \lim(a_n + b_n)$ and $\{b_n\}$ diverges. Then $b_n = (a_n + b_n) - a_n$ for all n . $\therefore s - a = \lim b_n$ (ALT), contradicting that $\{b_n\}$ diverges. $\square$

False; if $\{a_n\}$ and $\{a_n + b_n\}$ converge then $\{b_n\} = \{(a_n + b_n) - a_n\}$ converges, by the Algebraic Limit Theorem.

(f) **True.** $b_n := 1/n \neq 0$. Given $\varepsilon > 0$ take $N > 1/\varepsilon$ (Archimedean): for $n \geq N$, $|b_n - 0| = \frac{1}{n} \leq \frac{1}{N} < \varepsilon \therefore b_n \rightarrow 0$. $1/b_n = n$ is unbounded (as in (d)) $\therefore \{1/b_n\}$ diverges.

True; $b_n = 1/n \rightarrow 0$, $b_n \neq 0$, but $1/b_n = n$ diverges (unbounded).

(g) **False.**

Proof. Suppose $\{a_n\}$ is unbounded, $\{b_n\}$ converges, $\{a_n - b_n\}$ is bounded. Then there are $M_1, M_2 > 0$ with $|a_n - b_n| \leq M_1$ and $|b_n| \leq M_2$ for all n (convergent $\implies$ bounded).

$$|a_n| = |(a_n - b_n) + b_n| \leq M_1 + M_2 \quad \text{for all } n \in \mathbb{N}$$

$\therefore \{a_n\}$ is bounded, a contradiction. $\square$

False; $a_n = (a_n - b_n) + b_n$ is bounded whenever $\{a_n - b_n\}$ and $\{b_n\}$ are, by the triangle inequality.

(h) **True.** $a_n := 0$, $b_n := (-1)^n$. Suppose $b = \lim b_n$; take $\varepsilon = 1$: there is N with $|b_n - b| < 1$ for all $n \geq N$. Take even $n_1 \geq N$, odd $n_2 \geq N$:

$$2 = |b_{n_1} - b_{n_2}| \leq |b_{n_1} - b| + |b - b_{n_2}| < 2 \quad (\text{contradiction})$$

$\therefore \{b_n\}$ diverges; $a_n b_n = 0$ for all $n \therefore \{a_n b_n\}$ converges to 0.

True; $a_n = 0 (\rightarrow 0)$, $b_n = (-1)^n$ (diverges), $a_n b_n = 0 (\rightarrow 0)$.

Question 2 (ε - N verification of three limits)

Problem

Verify, using the definition of limits of sequences, that the following sequences converge to the proposed limits.

(a) $\frac{2}{5} = \lim_{n \rightarrow \infty} \frac{2n+1}{5n+4}.$

(b) $0 = \lim_{n \rightarrow \infty} \frac{2n^2}{n^3+3}.$

(c) $0 = \lim_{n \rightarrow \infty} \frac{\sin(n^2)}{\sqrt[3]{n}}.$

(Tutorial 2, Q2.)

Solution

Method 1: bound $|a_n - a|$ by C/n^p , invert for N

(a) $\frac{2}{5} = \lim_{n \rightarrow \infty} \frac{2n+1}{5n+4}.$

Proof. Let $\varepsilon > 0$; choose $N \in \mathbb{N}$ with $N > \frac{3}{25\varepsilon}$ (Archimedean). For all $n \geq N$:

$$\begin{aligned} \left| \frac{2n+1}{5n+4} - \frac{2}{5} \right| &= \frac{|5(2n+1) - 2(5n+4)|}{5(5n+4)} = \frac{3}{5(5n+4)} \\ &< \frac{3}{25n} \leq \frac{3}{25N} < \varepsilon \quad (5n+4 > 5n) \end{aligned}$$

$\therefore \frac{2}{5} = \lim_{n \rightarrow \infty} \frac{2n+1}{5n+4}$ (def. of limit). □

(b) $0 = \lim_{n \rightarrow \infty} \frac{2n^2}{n^3+3}.$

Proof. Let $\varepsilon > 0$; choose $N \in \mathbb{N}$ with $N > \frac{2}{\varepsilon}$ (Archimedean). For all $n \geq N$:

$$\left| \frac{2n^2}{n^3+3} - 0 \right| < \frac{2n^2}{n^3} = \frac{2}{n} \leq \frac{2}{N} < \varepsilon \quad (n^3+3 > n^3 > 0)$$

$\therefore 0 = \lim_{n \rightarrow \infty} \frac{2n^2}{n^3+3}$ (def. of limit). □

(c) $0 = \lim_{n \rightarrow \infty} \frac{\sin(n^2)}{\sqrt[3]{n}}.$

Proof. Let $\varepsilon > 0$; choose $N \in \mathbb{N}$ with $N > \frac{1}{\varepsilon^3}$ (Archimedean), so $\sqrt[3]{N} > \frac{1}{\varepsilon}$. For all $n \geq N$:

$$\left| \frac{\sin(n^2)}{\sqrt[3]{n}} - 0 \right| \leq \frac{1}{\sqrt[3]{n}} \leq \frac{1}{\sqrt[3]{N}} < \varepsilon \quad (|\sin(n^2)| \leq 1)$$

$\therefore 0 = \lim_{n \rightarrow \infty} \frac{\sin(n^2)}{\sqrt[3]{n}}$ (def. of limit). □

Question 3 (Bounded times null, and a warning when the limit is nonzero)

Problem

In this question, $\{a_n\}$ is a bounded sequence and $\{b_n\}$ is a convergent sequence.

- (a) Prove that if $\{b_n\}$ has limit 0, then the sequence $\{a_n b_n\}$ has limit 0.
- (b) Give an example to show that the sequence $\{a_n b_n\}$ may not have a limit if $\{b_n\}$ has a limit different from 0.

(Tutorial 2, Q3.)

Solution

Method 1: Squeeze Theorem; oscillating example

(a).

Proof. Let $M > 0$ with $|a_n| \leq M$ for all n . $\therefore -M|b_n| \leq a_n b_n \leq M|b_n|$ for all n .
 $||b_n| - 0| = |b_n - 0| \therefore |b_n| \rightarrow 0 \therefore \pm M|b_n| \rightarrow 0$ (ALT). $\therefore 0 = \lim a_n b_n$ (Squeeze). $\square$

$$0 = \lim a_n b_n \text{ whenever } \{a_n\} \text{ is bounded and } 0 = \lim b_n.$$

(b). $a_n := (-1)^n$ ($|a_n| \leq 1$), $b_n := 1 \rightarrow 1 \neq 0$; $a_n b_n = (-1)^n$. Suppose $L = \lim(-1)^n$; take $\varepsilon = 1$: there is N with $|(-1)^n - L| < 1$ for all $n \geq N$. Take even $n_1 \geq N$, odd $n_2 \geq N$:

$$2 = |1 - (-1)| \leq |1 - L| + |L - (-1)| < 2 \quad (\text{contradiction})$$

$\therefore \{a_n b_n\}$ diverges.

$$a_n = (-1)^n, b_n = 1 (\rightarrow 1 \neq 0), a_n b_n = (-1)^n \text{ diverges.}$$

Method 2: direct ε - N (part (a))

Let $M > 0$ with $|a_n| \leq M$, and $\varepsilon > 0$. Since $b_n \rightarrow 0$, there is N with $|b_n| < \varepsilon/M$ for all $n \geq N$.

$$\therefore |a_n b_n - 0| \leq M|b_n| < M \cdot \frac{\varepsilon}{M} = \varepsilon \quad \text{for all } n \geq N \implies 0 = \lim a_n b_n$$

Question 4 (Three ε - N proofs with the Algebraic Limit Theorem banned)

Problem

In this question, $\{a_n\}$ is a sequence of positive real numbers. Use the definition of limits of sequences to prove the following statements. **Do not use the Algebraic Limit Theorem.**

(a) If $2 = \lim a_n$, then $1 = \lim \frac{2a_n - 1}{3}$.

(b) If $2 = \lim a_n$, then $\sqrt{2} = \lim \sqrt{a_n}$.

(c) If $2 = \lim a_n$, then $\frac{1}{2} = \lim \frac{1}{a_n}$.

(Tutorial 2, Q4.)

Solution

Method 1: direct ε - N estimates

(a) $1 = \lim \frac{2a_n - 1}{3}$.

Proof. Let $\varepsilon > 0$. Since $2 = \lim a_n$, there is N with $|a_n - 2| < \frac{3\varepsilon}{2}$ for all $n \geq N$. For all $n \geq N$:

$$\left| \frac{2a_n - 1}{3} - 1 \right| = \frac{2}{3} |a_n - 2| < \varepsilon$$

$\therefore 1 = \lim \frac{2a_n - 1}{3}$ (def. of limit). □

$$1 = \lim \frac{2a_n - 1}{3}.$$

(b) $\sqrt{2} = \lim \sqrt{a_n}$.

Proof. Let $\varepsilon > 0$. Since $2 = \lim a_n$, there is N with $|a_n - 2| < \sqrt{2}\varepsilon$ for all $n \geq N$. For all $n \geq N$:

$$\left| \sqrt{a_n} - \sqrt{2} \right| = \frac{|a_n - 2|}{\sqrt{a_n} + \sqrt{2}} < \frac{|a_n - 2|}{\sqrt{2}} < \varepsilon \quad (\sqrt{a_n} > 0)$$

$\therefore \sqrt{2} = \lim \sqrt{a_n}$ (def. of limit). □

$$\sqrt{2} = \lim \sqrt{a_n}.$$

(c) $\frac{1}{2} = \lim \frac{1}{a_n}$.

Proof. Since $2 = \lim a_n$, there is N_1 with $|a_n - 2| < 1$ for all $n \geq N_1$. $\therefore a_n > 1$ for all $n \geq N_1$. Let $\varepsilon > 0$; there is N_2 with $|a_n - 2| < 2\varepsilon$ for all $n \geq N_2$. Put $N := \max\{N_1, N_2\}$. For all $n \geq N$:

$$\left| \frac{1}{a_n} - \frac{1}{2} \right| = \frac{|a_n - 2|}{2a_n} < \frac{|a_n - 2|}{2} < \varepsilon \quad (a_n > 1)$$

$$\therefore \frac{1}{2} = \lim_{n \rightarrow \infty} \frac{1}{a_n} \text{ (def. of limit).}$$

□

$$\boxed{\frac{1}{2} = \lim_{n \rightarrow \infty} \frac{1}{a_n}.}$$

Question 5 (Cauchy sequences: scalar multiples and squares, without the Cauchy Criterion)

Problem

In this question, $\{a_n\}$ is a Cauchy sequence and c is a real number.

- (a) Use the definition of Cauchy sequence to show that $\{ca_n\}$ is Cauchy. **Do not use the Cauchy Criterion or the Algebraic Limit Theorem.**
- (b) Use the definition of Cauchy sequence to show that $\{a_n^2\}$ is Cauchy. **Do not use the Cauchy Criterion or the Algebraic Limit Theorem.**

[Note: A proof without using the Cauchy Criterion is applicable to sequences in $\mathbb{Q}$ (or any ordered field) for which the Cauchy Criterion does not hold.] (*Tutorial 2, Q5; part (b) is also the sample midterm Q4.*)

Solution

Method 1: definition-only Cauchy estimates

(a) $\{ca_n\}$ is Cauchy.

Proof. Case $c = 0$: $|ca_n - ca_m| = 0 < \varepsilon$ for all n, m and every $\varepsilon > 0$ $\therefore \{ca_n\}$ is Cauchy.
Case $c \neq 0$: let $\varepsilon > 0$. There is N with $|a_n - a_m| < \varepsilon/|c|$ for all $n, m \geq N$. For all $n, m \geq N$:

$$|ca_n - ca_m| = |c| |a_n - a_m| < \varepsilon$$

$\therefore \{ca_n\}$ is Cauchy (def. of Cauchy). □

$\{ca_n\} \text{ is Cauchy, for every } c \in \mathbb{R}.$

(b) $\{a_n^2\}$ is Cauchy.

Proof. Step 1: $\{a_n\}$ is bounded. Take $\varepsilon = 1$: there is N_0 with $|a_n - a_m| < 1$ for all $n, m \geq N_0$. For all $n \geq N_0$:

$$|a_n| \leq |a_n - a_{N_0}| + |a_{N_0}| < 1 + |a_{N_0}| \quad (m = N_0)$$

$M := \max\{|a_1|, \dots, |a_{N_0-1}|, 1 + |a_{N_0}|\} > 0$ $\therefore |a_n| \leq M$ for all $n \in \mathbb{N}$.

Step 2. Let $\varepsilon > 0$. There is N with $|a_n - a_m| < \frac{\varepsilon}{2M}$ for all $n, m \geq N$. For all $n, m \geq N$:

$$\begin{aligned} |a_n^2 - a_m^2| &= |a_n - a_m| |a_n + a_m| \\ &\leq |a_n - a_m| \cdot 2M < \varepsilon \end{aligned} \quad (|a_n| + |a_m| \leq 2M)$$

$\therefore \{a_n^2\}$ is Cauchy (def. of Cauchy). □

$\{a_n^2\}$ is Cauchy (boundedness of $\{a_n\}$, proved directly from the Cauchy definition, is the key extra ingredient).

Question A1 (A linear recurrence $b_n = a_{n+1} + ra_n$ forces $\{a_n\}$ to converge)

Problem

Prove that for any sequence $\{a_n\}$ and any real number r with $|r| < 1$, if the sequence $\{b_n := a_{n+1} + ra_n\}$ is convergent, then the sequence $\{a_n\}$ is convergent. (*Tutorial 2, Additional Question A1.*)

Solution

Method 1: inductive geometric-decay estimate

Proof. Let $\rho := |r| < 1$, $b := \lim b_n$, $a := \frac{b}{1+r}$ ($1+r > 0$), $e_n := a_n - a$.

$$\begin{aligned} b_n - b &= (a_{n+1} - a) + r(a_n - a) & (b = (1+r)a) \\ \implies e_{n+1} &= (b_n - b) - re_n \implies |e_{n+1}| \leq \rho |e_n| + |b_n - b| \quad (n \in \mathbb{N}) \end{aligned}$$

Lemma: $\rho^k \rightarrow 0$. Trivial if $\rho = 0$. Else $\rho = \frac{1}{1+h}$, $h := \frac{1}{\rho} - 1 > 0$, and $(1+h)^k \geq 1+kh > kh$ (Bernoulli). Let $\varepsilon > 0$. $k > \frac{1}{h\varepsilon}$ (Archimedean) $\implies \rho^k < \frac{1}{kh} < \varepsilon$.

Step 1. Fix $\eta > 0$ and N_0 with $|b_n - b| < \eta$ for all $n \geq N_0$. Claim, for all $k \in \mathbb{N}_0$:

$$|e_{N_0+k}| \leq \rho^k |e_{N_0}| + \frac{\eta}{1-\rho}$$

$k = 0$: clear. Step $k \rightarrow k+1$:

$$\begin{aligned} |e_{N_0+k+1}| &\leq \rho |e_{N_0+k}| + \eta \leq \rho^{k+1} |e_{N_0}| + \frac{\rho\eta}{1-\rho} + \eta \quad (\text{induction hypothesis}) \\ &= \rho^{k+1} |e_{N_0}| + \frac{\eta}{1-\rho} \end{aligned}$$

Step 2. Let $\varepsilon > 0$; take $\eta := \frac{1}{2}\varepsilon(1-\rho)$ and N_0 as in Step 1. By the Lemma (with $\varepsilon/(2|e_{N_0}|+1)$) choose K with $\rho^K |e_{N_0}| < \frac{\varepsilon}{2}$. Put $N := N_0 + K$. For $n \geq N$, $k := n - N_0 \geq K$ and $\rho^k \leq \rho^K$:

$$|a_n - a| = |e_{N_0+k}| \leq \rho^K |e_{N_0}| + \frac{\varepsilon}{2} < \varepsilon \quad (\rho^k \leq \rho^K, \frac{\eta}{1-\rho} = \frac{\varepsilon}{2})$$

$\therefore a = \lim a_n$ (def. of limit). □

$\{a_n\} \text{ converges, with } \lim a_n = \frac{\lim b_n}{1+r}.$

Question A2 (Cesaro means of a convergent sequence)

Problem

Show that if the sequence $\{a_n\}$ has the limit a , then the sequence $\{b_n\}$ with

$$b_n = \frac{a_1 + a_2 + \cdots + a_n}{n}$$

for every $n \in \mathbb{N}$ converges to a . (*Tutorial 2, Additional Question A2.*)

Solution

Method 1: bounded head, $\varepsilon/2$ -tail

Proof. Convergent $\implies$ bounded: $|a_n| \leq C$ for all $n \therefore |a_n - a| \leq C + |a| =: C'$ for all n . Let $\varepsilon > 0$. There is M with $|a_k - a| < \frac{\varepsilon}{2}$ for all $k > M$. Choose $N > M$ with $N > \frac{2MC'}{\varepsilon}$ (Archimedean). For $n \geq N$:

$$\begin{aligned} |b_n - a| &= \left| \frac{1}{n} \sum_{k=1}^n (a_k - a) \right| \leq \frac{1}{n} \sum_{k=1}^M |a_k - a| + \frac{1}{n} \sum_{k=M+1}^n |a_k - a| \\ &\leq \frac{MC'}{n} + \frac{n-M}{n} \cdot \frac{\varepsilon}{2} < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon \end{aligned} \quad (n > \frac{2MC'}{\varepsilon})$$

$\therefore a = \lim b_n$ (def. of limit). □

$$a = \lim b_n, \quad \text{where } b_n = \frac{a_1 + \cdots + a_n}{n}.$$

Question A3 (Doubly indexed arrays: iterated limits versus the double limit)

Problem

Given a doubly indexed array a_{mn} where $m, n \in \mathbb{N}$, define $a = \lim_{m,n \rightarrow \infty} a_{mn}$ to mean

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall m, n \geq N \quad |a_{mn} - a| < \varepsilon.$$

[Note: a doubly indexed array is a function $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ where we use a_{mn} to denote $f(m, n)$.]

- If the array satisfies $a_{mn} = m/(m+n)$ for each $m, n \in \mathbb{N}$, what are the iterated limits $\lim_{n \rightarrow \infty} (\lim_{m \rightarrow \infty} a_{mn})$ and $\lim_{m \rightarrow \infty} (\lim_{n \rightarrow \infty} a_{mn})$?
- If the array satisfies $a_{mn} = mn/(m+n)$ for each $m, n \in \mathbb{N}$, does there exist $a \in \mathbb{R}$ for which $a = \lim_{m,n \rightarrow \infty} a_{mn}$? Do the two iterated limits exist? How do these three values compare?
- Prove that if $a = \lim_{m,n \rightarrow \infty} a_{mn}$ and the iterated limit $\lim_{n \rightarrow \infty} (\lim_{m \rightarrow \infty} a_{mn})$ exists, then the iterated limit is a .

(Tutorial 2, Additional Question A3.)

Solution

Method 1: single-variable ε - N for each inner limit

(a) $a_{mn} = m/(m+n)$.

Proof. Fix n . For all m : $\left| \frac{m}{m+n} - 1 \right| = \frac{n}{m+n} < \frac{n}{m}$. Given $\varepsilon > 0$, take $M > n/\varepsilon$ (Archimedean): $m \geq M \implies |a_{mn} - 1| < \frac{n}{M} < \varepsilon \therefore \lim_{m \rightarrow \infty} a_{mn} = 1$ for every n . Fix m . For all n : $\left| \frac{m}{m+n} - 0 \right| < \frac{m}{n}$. Given $\varepsilon > 0$, take $N > m/\varepsilon$: $n \geq N \implies |a_{mn}| < \frac{m}{N} < \varepsilon \therefore \lim_{n \rightarrow \infty} a_{mn} = 0$ for every m . $\therefore \lim_{n \rightarrow \infty} (\lim_{m \rightarrow \infty} a_{mn}) = \lim_{n \rightarrow \infty} 1 = 1$ and $\lim_{m \rightarrow \infty} (\lim_{n \rightarrow \infty} a_{mn}) = \lim_{m \rightarrow \infty} 0 = 0$ (constant sequences). $\square$

$$\lim_{n \rightarrow \infty} \left(\lim_{m \rightarrow \infty} a_{mn} \right) = 1.$$

$$\lim_{m \rightarrow \infty} \left(\lim_{n \rightarrow \infty} a_{mn} \right) = 0.$$

(b) $a_{mn} = mn/(m+n)$.

Proof: no double limit. Suppose $a = \lim_{m,n \rightarrow \infty} a_{mn}$; take $\varepsilon = 1$: there is N with $a_{mn} < a + 1$ for all $m, n \geq N$. Choose $n_0 \geq N$ with $n_0 > 2(a+1)$ (Archimedean). Then $a_{n_0 n_0} = \frac{n_0}{2} > a + 1$, a contradiction. $\square$

Proof: no iterated limit. Fix n . For all m : $\left| \frac{mn}{m+n} - n \right| = \frac{n^2}{m+n} < \frac{n^2}{m}$. Given $\varepsilon > 0$, take $M > n^2/\varepsilon$: $m \geq M \implies |a_{mn} - n| < \varepsilon \therefore \lim_{m \rightarrow \infty} a_{mn} = n$. $\{n\}$ is unbounded (Archimedean) $\therefore \lim_{n \rightarrow \infty} (\lim_{m \rightarrow \infty} a_{mn}) = \lim_{n \rightarrow \infty} n$ does not exist (convergent $\implies$ bounded). $a_{mn} = a_{nm} \therefore \lim_{n \rightarrow \infty} a_{mn} = m$ and $\lim_{m \rightarrow \infty} m$ does not exist either. $\square$

As printed ($a_{mn} = mn/(m+n)$): the double limit and both iterated limits all fail to exist.

(b') $a_{mn} = mn/(m^2 + n^2)$.

Proof: no double limit. $a_{nn} = \frac{1}{2}$ and $a_{2n,n} = \frac{2}{5}$ for all n . Suppose $a = \lim_{m,n \rightarrow \infty} a_{mn}$; take $\varepsilon = \frac{1}{20}$ and N as in the definition. (N, N) and $(2N, N)$ have both entries $\geq N$ $\therefore \left| \frac{1}{2} - a \right| < \varepsilon$ and $\left| \frac{2}{5} - a \right| < \varepsilon$.

$$\frac{1}{10} = \left| \frac{1}{2} - \frac{2}{5} \right| \leq \left| \frac{1}{2} - a \right| + \left| a - \frac{2}{5} \right| < \frac{1}{10} \quad (\text{contradiction})$$

$\square$

Proof: both iterated limits are 0. Fix n . $0 < a_{mn} \leq \frac{mn}{m^2} = \frac{n}{m}$ because $m^2 + n^2 \geq m^2$. Given $\varepsilon > 0$, take $M > n/\varepsilon$: $m \geq M \implies |a_{mn}| < \varepsilon \therefore \lim_{m \rightarrow \infty} a_{mn} = 0$ for every n $\therefore \lim_{n \rightarrow \infty} (\lim_{m \rightarrow \infty} a_{mn}) = 0$. $a_{mn} = a_{nm} \therefore \lim_{m \rightarrow \infty} (\lim_{n \rightarrow \infty} a_{mn}) = 0$. $\square$

$a_{mn} = mn/(m^2 + n^2)$: double limit does not exist; both iterated limits equal 0.

(c).

Proof. Let $b_n := \lim_{m \rightarrow \infty} a_{mn}$ and $\varepsilon > 0$. There is N_1 with $|a_{mn} - a| < \frac{\varepsilon}{2}$ for all $m, n \geq N_1$. Fix $n \geq N_1$. Since $a_{mn} \rightarrow b_n$ as $m \rightarrow \infty$, there is $m_0 \geq N_1$ with $|a_{m_0 n} - b_n| < \frac{\varepsilon}{2}$.

$$|b_n - a| \leq |b_n - a_{m_0 n}| + |a_{m_0 n} - a| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

$\therefore |b_n - a| < \varepsilon$ for all $n \geq N_1 \therefore a = \lim_{n \rightarrow \infty} b_n$, i.e. the iterated limit is a . $\square$

If $a = \lim_{m,n \rightarrow \infty} a_{mn}$ and $\lim_{n \rightarrow \infty} (\lim_{m \rightarrow \infty} a_{mn})$ exists, it equals a .

Method Map

Q.	Core topic	Methods
1	True/false on convergence	Explicit examples; contradiction from the definition, ALT, boundedness, triangle inequality.
2	ε - N verification of three limits	Bound by C/n^p , invert with the Archimedean Property.
3	Bounded $\times$ null; nonzero-limit warning	Squeeze Theorem; direct ε - N (Method 2); counterexample $(-1)^n$.
4	Three definition-only proofs	Direct ε - N ; conjugate for $\sqrt{a_n}$; two-stage bound for $1/a_n$.
5	Cauchy: scalar multiple, square	Case split on c ; Cauchy $\implies$ bounded from the definition; $ a_n^2 - a_m^2 \leq 2M a_n - a_m $.
A1	Recurrence $b_n = a_{n+1} + ra_n$	Induction on $ e_{N_0+k} $; Bernoulli for $ r ^k \rightarrow 0$.
A2	Cesaro means	Bounded head plus $\varepsilon/2$ -tail.
A3	Double vs. iterated limits	Single-variable ε - N ; triangle inequality in (c).

Final Answers

- (a) True, (b) False, (c) True, (d) True, (e) False, (f) True, (g) False, (h) True.
- $\frac{2}{5} = \lim \frac{2n+1}{5n+4}$, $0 = \lim \frac{2n^2}{n^3+3}$, $0 = \lim \frac{\sin(n^2)}{\sqrt[3]{n}}$; any $N \in \mathbb{N}$ with $N > 3/(25\varepsilon)$, $N > 2/\varepsilon$, $N > 1/\varepsilon^3$ respectively.
- (a) $0 = \lim a_n b_n$. (b) $a_n = (-1)^n$, $b_n = 1$ gives $a_n b_n = (-1)^n$, divergent.
- (a) $1 = \lim \frac{2a_n - 1}{3}$. (b) $\sqrt{2} = \lim \sqrt{a_n}$. (c) $\frac{1}{2} = \lim \frac{1}{a_n}$.
- (a) $\{ca_n\}$ is Cauchy for every $c \in \mathbb{R}$. (b) $\{a_n^2\}$ is Cauchy.
- $\{a_n\}$ converges, with $\lim a_n = (\lim b_n)/(1+r)$.
- $a = \lim b_n$, where $b_n = (a_1 + \cdots + a_n)/n$.
- (a) $\lim_n(\lim_m a_{mn}) = 1$, $\lim_m(\lim_n a_{mn}) = 0$. (b) $mn/(m+n)$: double limit and both iterated limits fail to exist; $mn/(m^2+n^2)$: double limit fails to exist, both iterated limits equal 0. (c) The iterated limit, when it exists, equals a .